

CONTACT

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PROFILE

AI/ML Technical Lead and Data Scientist with 7+ years building and scaling production ML systems. I design end-to-end ML products — from problem framing and experimentation to deployment and monitoring — that move real business KPIs. Deep in causal measurement and A/B testing, uplift and propensity modeling, recommendation systems, and applied LLMs.

EXPERTISE

AI & ML: Machine Learning, Deep Learning, Generative AI & LLMs, NLP, Embeddings, Recommendation & Ranking Systems, Graph ML, Computer Vision.

ML Engineering: End-to-end ML systems, model deployment & serving, batch + real-time inference, feature engineering, experiment tracking, model monitoring & drift, MLOps / LLMOps.

Experimentation & Modeling: A/B testing, causal inference, uplift & propensity modeling, forecasting, classification, optimization, anomaly detection.

Programming & Cloud: Python, Go, R, SQL, BigQuery, Redis, MySQL, GCP, AWS, Git, Linux.

Frameworks: Scikit-Learn, LightGBM, XGBoost, scikit-learn, PyTorch, TensorFlow, Keras, Pandas, NumPy, Optuna, Flask, Streamlit, Plotly.

EDUCATION

M.Sc. in Analytical Intelligence for Decision Making
Universidad de los Andes, Colombia
Aug 2019 - Apr 2021

B.Sc. Economics
Politécnico Grancolombiano, Colombia
Aug 2014 - Mar 2019

SERGIO A. MORA PARDO

AI / ML TECHNICAL LEAD & DATA SCIENTIST
Production ML at scale · Causal experimentation · Recommendation & ranking · Applied LLMs & evaluation

WORK EXPERIENCE

Data Scientist Technical Leader

Mercado Libre · Jul 2024 - Jul 2026

- Led ranking and discovery ML for the Affiliates marketplace: designed and shipped an A/B test (~5.2M sessions, 4 countries) replacing a static curated landing with a continuous, personalized product-discovery feed — lifting engagement +20–26% across the funnel (clicks, intent, shares) and +17% conversion on newly-generated affiliate links vs. control.
- Built the affiliate attribution and product-scoring pipeline — resolving shared links to orders across web and app — creating the analytical source of truth for affiliate sales measurement, purchase-propensity and price ranking signals, and controlled A/B experiments across placements (hub and product page).
- Led applied research on LLM systems for personalized recommendations: built an offline evaluation framework with ground-truth harnesses, precision/recall@K and bootstrap CIs for candidate retrieval, and guided the technical design of an MCP-based (Model Context Protocol) tool layer feeding LLM-generated personalized push notifications — establishing the team's LLM evaluation methodology.
- As technical lead, reviewed 200+ pull requests from ~18 engineers across 6 repositories and 5 countries (Brazil, Argentina, Uruguay, Colombia, Chile), driving code quality, experiment design and ML/AI best practices.

Senior Data Scientist

Mercado Libre · Feb 2022 - Jul 2024

- Built and led the ML remarketing engine (push & email) across 8 countries — causal uplift (LightGBM / scikit-learn), open-propensity audience filtering (XGBoost) and send-time (moment) models — co-designing 38+ factorial (2x2) A/B experiments and driving ~US\$1M/month in statistically significant incremental GMV attributable to ML optimizations vs. holdout control.
- Designed the production serving architecture: batch scoring in BigQuery loaded into a low-latency key-value store, served by prediction APIs and a real-time event-streaming session engine that decides audience, timing and content — serving personalized push & email to ~12M monthly users across Brazil, Mexico, Argentina, Colombia, Chile, Peru, Uruguay and Ecuador.
- Drove double-digit conversion lift vs. control: ML audience filtering with the open/propensity model +25% and price-aware recommendation ranking +19% on recommended-item conversion — while rigorously retiring underperformers (trigger-based -0.4%).
- Framed and delivered recurring campaigns — remarketing, cross-sell and buy-again — using RFM segmentation and purchase-propensity models to drive activation and repeat purchase.

Adjunct Professor

Pontificia Universidad Javeriana · Aug 2023 - Present

- Design and teach two graduate & specialization courses — Deep Learning, NLP, Graph ML & Computer Vision and Stochastic AI Techniques (classical ML, time series, Monte Carlo methods, anomaly detection) — to ~20 students per cohort across 5+ cohorts since 2023.
- Built and modernized the curriculum from scratch, with hands-on projects in PyTorch / TensorFlow (CNNs, transformers/BERT, LSTMs, graph networks, LLMs) and Kaggle-style competitions, earning a P97 (top 3%) teaching-evaluation score.

Director Advanced Analytics & BI

Politécnico Grancolombiano · Apr 2021 - Feb 2022

- Led a 6-person Analytics & Data team and the institution's analytics/ML platform and delivery portfolio, providing descriptive, predictive and prescriptive solutions through APIs, dashboards and business workflows.

Analytics & Data Leader - Lead Data Scientist

Politécnico Grancolombiano · Aug 2020 - Apr 2021

- Worked as a hands-on individual contributor across the end-to-end ML lifecycle - problem framing, modeling, deployment and monitoring - building systems with ensemble and linear models, RNN/CNN, NLP transfer learning and optimization, served through Flask REST APIs and Streamlit/Dash/Shiny dashboards on AWS.

PUBLICATIONS

- Light Innovation in Energy Supply for Non-connected Areas in Colombia: Partial Research via the Forgotten Effects Model. In Modelling and Simulation in Management Sciences — Advances in Intelligent Systems and Computing, Vol. 894, Springer (2019), pp. 35–50. Co-author. doi.org/10.1007/978-3-030-15413-4_4
- Sistemas de Innovación en Latinoamérica: una mirada compartida desde México, Colombia y Chile. Real Academia de Ciencias Económicas y Financieras, Barcelona (2019). ISBN 978-84-09-08037-3. Co-author. racef.es
- El sector energético colombiano: las zonas no interconectadas vistas a través de la innovación ligera. Working Papers MGEM 2(1), Politécnico Grancolombiano (2017). Co-author. doi.org/10.15765/wpmgem.v2i1.971
- Red Iberoamericana para la Competitividad, Innovación y Desarrollo. Working Papers MGEM 2(1), Politécnico Grancolombiano (2017). Co-author. doi.org/10.15765/wpmgem.v2i1.972